

Assignment 1 - Univariate and Multivariate Statistics

FinTech 545 - Quantitative Risk Management

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Written response

1. Reading the Shape of a Sample

Predict

Part (a).

The first four sample moments are:

Moment	Estimate
Mean	0.00110439
Sample variance	0.0000962482
Standardized skewness	-0.669837
Excess kurtosis	2.357614

The negative skewness indicates asymmetry toward the left, while the positive excess kurtosis indicates heavier tails than a Normal distribution. A Normal distribution is inconsistent with both features because it has zero skewness and zero excess kurtosis. A symmetric Student's t distribution can account for the positive excess kurtosis, but not the negative skewness. A lognormal distribution is also inconsistent with the sample because its skewness is positive. The normal inverse Gaussian (NIG) distribution remains plausible because it can accommodate both skewness and heavy tails.

Fit

Part (b).

I fit a Normal distribution by matching the sample mean and sample variance:

$$X \sim N(0.00110439, 0.0000962482).$$

The fitted standard deviation is 0.00981062, which gives a 1% quantile of -0.02171852. There are 26 observations below this threshold. Under the fitted Normal, the expected count is

$$1000(0.01)=10.$$

The empirical proportion below the fitted 1% quantile is therefore 2.6%.

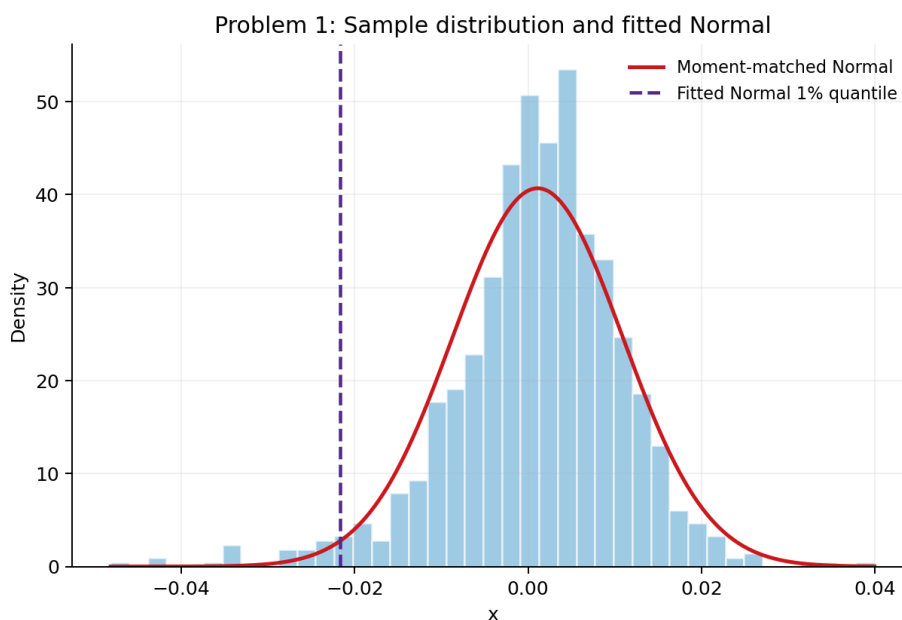


Figure 1. The sample distribution, moment-matched Normal density, and fitted Normal 1% quantile.

Reconcile

Part (c).

The fitted Normal places too little probability in the left tail. A threshold that should be crossed about 10 times is crossed 26 times. This agrees with the negative skewness and positive excess kurtosis seen before fitting: the Normal model understates downside tail risk for this sample.

2. A Regression Whose Errors Are Not Normal

Predict

Parts (a) and (b).

The scatter is roughly linear with a strong positive relationship, but a few observations sit much farther from the main pattern than I would expect under a Normal error. I therefore expect a heavy-tailed error distribution, such as Student's t, to fit better. This violates OLS assumption 7, Normal errors. Because OLS gives large residuals substantial weight, I also expect the unusual observations to move the slope somewhat. I expect the large residuals to increase the estimated residual variance and make the conventional standard error of the slope larger. Normal-based inference may also be less reliable.

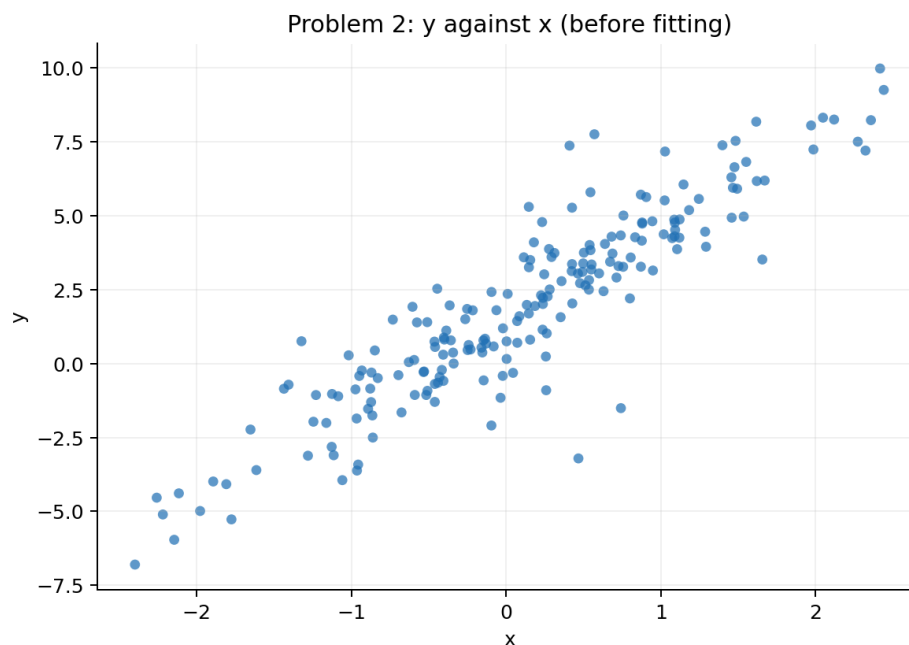


Figure 2. The y-versus-x scatter viewed before fitting any regression model.

Fit

I estimated OLS, a Normal-error maximum-likelihood regression, and a Student's-t-error maximum-likelihood regression. The Student's t model uses the parameterization

$$\varepsilon \sim t(v, \text{location}=0, \text{scale}=s),$$

where the fitted scale is not the standard deviation.

Model	Alpha	Beta	Error parameters	Log likelihood	AICc
OLS	1.560740	2.990757	SE(alpha) = 0.096269; SE(beta) = 0.096640	-	-
Normal MLE	1.560740	2.990757	sigma = 1.344627	-343.011037	692.144523
Student's t MLE	1.547979	3.019179	scale = 0.935509; nu = 3.632165	-328.561179	665.327487

For the Normal model, the AICc parameter count is 3: alpha, beta, and sigma. For the Student's t model, it is 4: alpha, beta, scale, and nu. I calculated AICc as

$$AICc = 2k - 2 \log L + (2k^2 + 2k) / (n - k - 1)$$

The Student's t model lowers AICc by 26.817036 points.

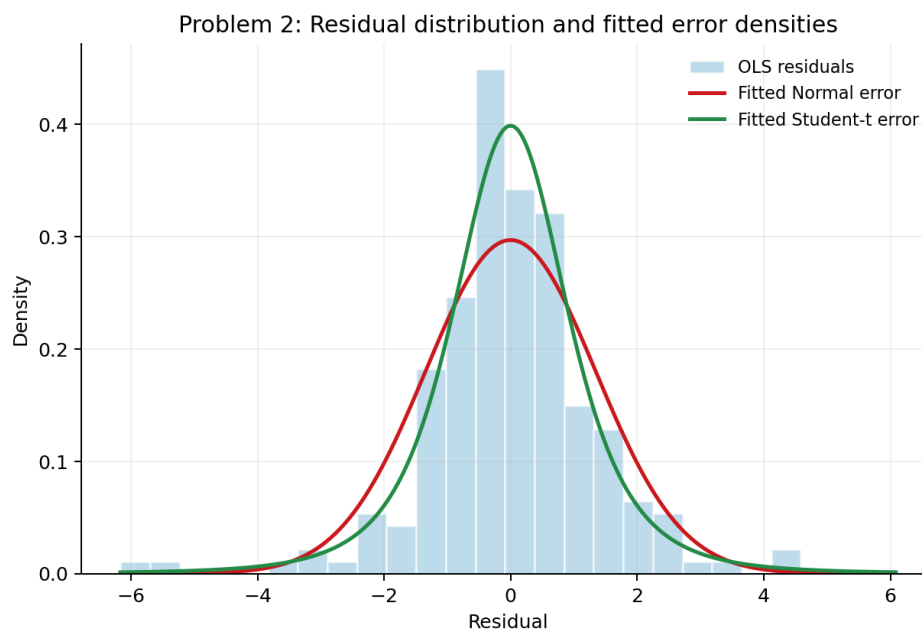


Figure 3. OLS residuals with the fitted Normal and Student's t error densities.

Reconcile

Part (c).

My prediction overstated the effect of the unusual observations on the slope. The OLS slope is 2.99076 and the Student's t slope is 3.01918, a difference of only about 0.0284. The central regression relationship barely changes.

Part (d). The rejected Normal model is getting the residual tails wrong, not the central slope. The Student's t likelihood assigns more probability to the unusually large errors, which is why its AICc is substantially lower.

Part (e).

The fitted positive error quantiles are:

Quantile	Normal error	Student's t error
95%	2.211715	2.053833
99.5%	3.463530	4.620085

The Student's t fit has a smaller scale and more mass near the center, so its 95% quantile is slightly below the Normal quantile. Farther into the tail, its low fitted degrees of freedom dominate, making the 99.5% quantile much larger. For a capital buffer, I would use the Student's t model because the extreme tail is the relevant risk.

3. Pearson Against Spearman

Predict

Part (a).

The x_1/x_2 scatter is strongly monotonic but visibly nonlinear, with an approximately cubic shape. I expect that pair to produce the largest Pearson-Spearman disagreement. The x_1/x_3 relationship looks closer to linear, and the pairs involving x_4 show little systematic association, so I expect the two measures to agree more closely for those pairs. The x_2/x_3 plot also appears monotonic but somewhat nonlinear, so I expect some disagreement there, although less than for x_1/x_2 .

Problem 3: Pairwise scatter plots (before correlations)

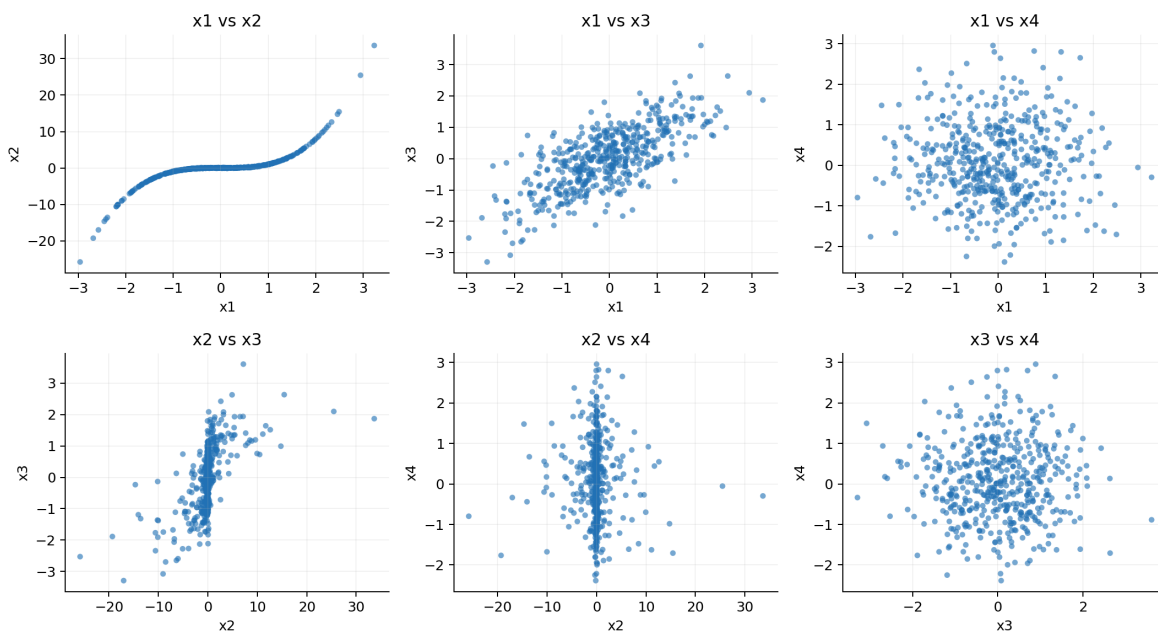


Figure 4. All pairwise scatter plots, viewed before calculating either correlation matrix.

Fit

Part (b).

The Pearson correlation matrix is:

	x1	x2	x3	x4
x1	1.000000	0.799092	0.719467	-0.004680
x2	0.799092	1.000000	0.587614	-0.020923
x3	0.719467	0.587614	1.000000	-0.008605
x4	-0.004680	-0.020923	-0.008605	1.000000

The Spearman correlation matrix is:

	x1	x2	x3	x4
x1	1.000000	0.971930	0.683910	0.000189
x2	0.971930	1.000000	0.667422	-0.001131
x3	0.683910	0.667422	1.000000	-0.002422
x4	0.000189	-0.001131	-0.002422	1.000000

The largest absolute gap is for x_1/x_2 :

$$|0.799092 - 0.971930| = 0.172838.$$

Reconcile

Part (c).

The result matches the shape visible in the pair plot. Spearman works with ranks, so a strong monotonic relationship can produce a value near one even when the curve is nonlinear. Pearson measures linear association and is lower because a straight line does not fully describe the cubic shape. Spearman is the more direct description if the question is whether x_2 moves monotonically with x_1 . Pearson answers the different question of how strong their linear association is. Neither coefficient is intrinsically wrong.

4. Conditional Distributions

Predict

Part (a).

Using the $n-1$ sample covariance convention, the covariance matrix for (X_1, X_2) is

$$\Sigma = \begin{bmatrix} 1.099875 & 1.696902 \\ 1.696902 & 4.104749 \end{bmatrix}$$

Here, $\Sigma_{11} = \text{Var}(X_1)$, $\Sigma_{22} = \text{Var}(X_2)$, and $\Sigma_{21} = \Sigma_{12} = \text{Cov}(X_2, X_1)$. Under the multivariate Normal model,

$$\text{Var}(X_2 | X_1) = \Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}.$$

For this sample, the conditional variance is 1.486746. The fraction of the original variance that remains is

$$(\Sigma_{22} - \Sigma_{21} \Sigma_{11}^{-1} \Sigma_{12}) / \Sigma_{22} = 0.362201$$

Learning X_1 therefore reduces the variance by about 63.78%.

Part (b). The factor is constant under the multivariate Normal model. The conditional variance formula contains no observed x_1 term, which settles the question before examining coverage in the data.

Fit

Part (c).

The conditional mean formula is

$$E[X_2 | X_1 = x_1] = \mu_2 + (\Sigma_{21} / \Sigma_{11})(x_1 - \mu_1)$$

The sample means are $\mu_1 = -0.016714$ and $\mu_2 = 0.045288$, and

$$\Sigma_{21} / \Sigma_{11} = 1.542813$$

Thus the fitted linear expression is

$$\text{Estimated } E[X_2 | X_1 = x_1] = 0.071073 + 1.542813x_1.$$

The coefficient 1.542813 is also the OLS slope from regressing x_2 on x_1 . The conditional standard deviation is 1.219322, so the constant-width 95% band is the fitted line plus or minus $1.96(1.219322)$.

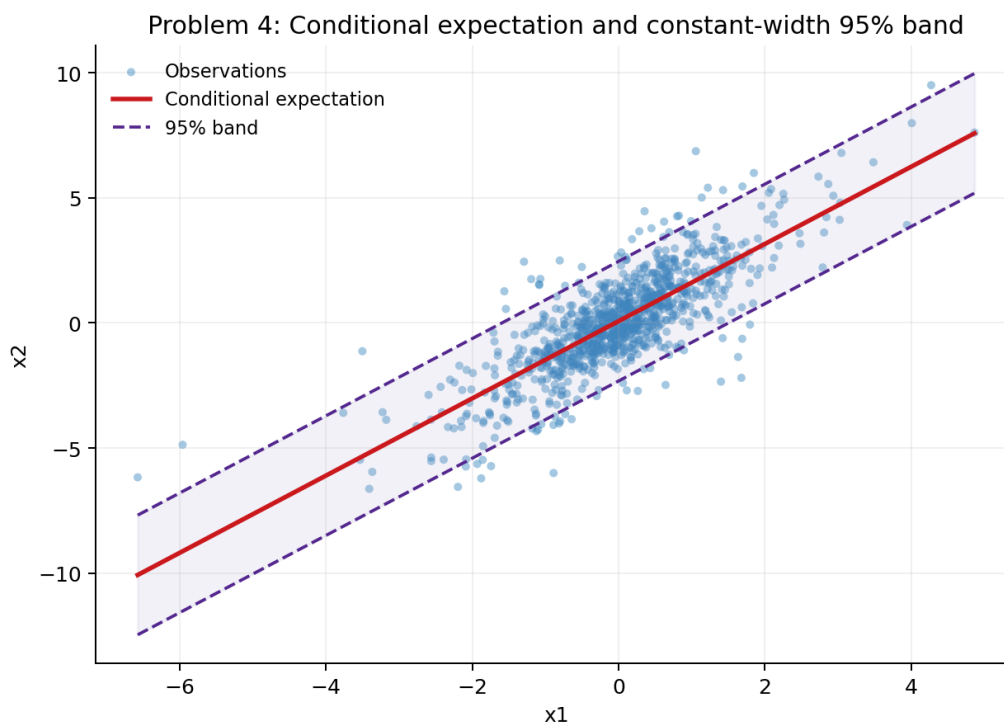


Figure 5. Conditional linear expression with the multivariate-Normal constant-width 95% band.

Parts (d) and (e).

Overall, 935 of the 1,000 observations lie inside the band, for 93.5% coverage. Splitting the sample by the distance of x_1 from its mean gives:

x_1 distance from its mean	Observations	Inside band	Coverage
At most 1 sample SD	759	726	95.652174%
More than 1 and at most 2 sample SDs	190	163	85.789474%
More than 2 sample SDs	51	46	90.196078%

Reconcile

Part (f).

The coverage is not flat across the three buckets. In particular, it falls well below 95% in the middle bucket. This indicates that the constant conditional variance implied by joint Normality is not a good description of the data.

The covariance-based coefficient still survives as the coefficient of the best linear predictor. Once joint Normality fails, however, the fitted linear expression is not guaranteed to be the exact conditional expectation, and the constant-width Normal band is not guaranteed to provide 95% conditional coverage at every value of x_1 .

5. Identifying an AR or MA Order

Predict

Parts (a) and (b).

The time-series plot appears stationary around a stable level. The ACF decays and oscillates rather than cutting off after a fixed lag. The PACF has strong spikes at lags 1 and 2, followed mostly by values inside the significance band

$$\pm 1.96 / \sqrt{500} = \pm 0.08765$$

There is an isolated significant PACF spike near lag 9, but it is not part of a persistent cutoff pattern. Using the rule that an AR(p) process has a decaying ACF and a PACF that cuts off after lag p, my pre-fit choice is AR(2).

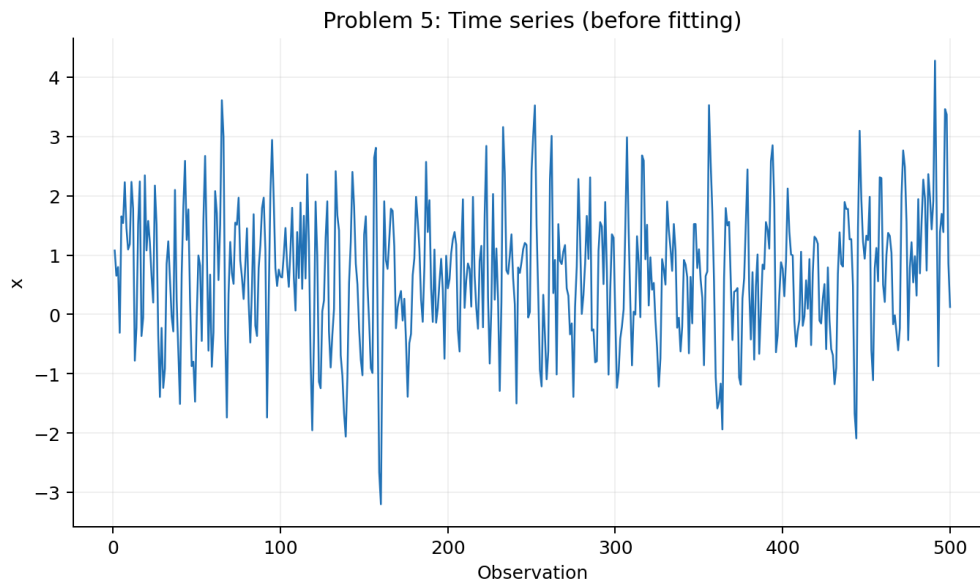


Figure 6. The time series viewed before fitting candidate AR and MA models.

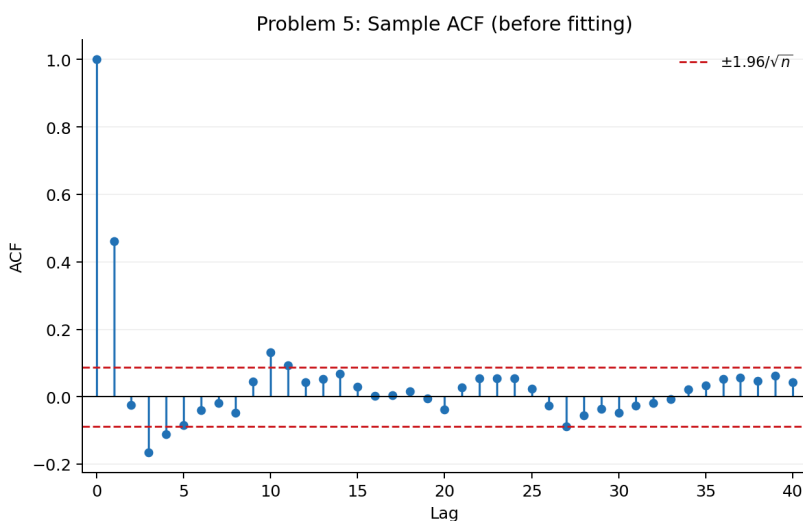


Figure 7. Sample ACF with the constant plus-or-minus 0.08765 significance band.

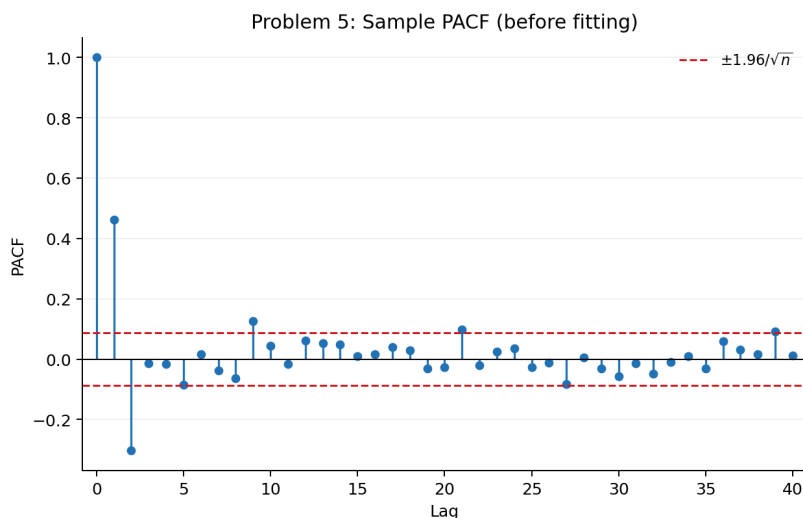


Figure 8. Sample PACF with the same significance band.

Fit

Part (c).

All six models include the same constant-mean treatment and use the same exact Gaussian state-space likelihood convention. The manually calculated AICc values are:

Model	AICc
AR(1)	1418.586959
AR(2)	1372.438310
AR(3)	1374.343211
MA(1)	1389.084281
MA(2)	1381.231619
MA(3)	1374.116525

The minimum is AR(2). Its fitted coefficients are

$$\phi_1 = 0.601201, \phi_2 = -0.302638.$$

For comparison, AR(3) estimates

$$\phi_1 = 0.596204, \phi_2 = -0.292690, \phi_3 = -0.016542.$$

The AR(2) and AR(3) log likelihoods are -682.178751 and -682.110877, respectively. Their AICc values are 1372.438310 and 1374.343211.

Reconcile

Part (d).

The AICc selection agrees with the AR(2) choice made from the ACF and PACF.

Part (e).

The third AR coefficient is only about -0.0165, and adding it produces a very small likelihood improvement. AICc rejects AR(3) because that gain does not justify the additional parameter. Ordinary R^2 does not impose the same complexity penalty. It cannot decrease when another regressor is added, so it would not make the same tradeoff between the small in-sample improvement and the extra parameter.